

		Therefore C is also wrong and centre of gravity is slightly to the left side of C.
7	C	Since the balloon is in equilibrium, upthrust on balloon = weight of balloon and helium + force by spring $\rho_{air}V_{balloon}g = \rho_{helium}V_{balloon}g + m_{balloon}g + kx$ $(1.29)(5.0)g = (0.180)(5.0)g + \left(\frac{3.5}{1000}\right)g + (100)x$ $x = 0.544 \text{ m}$
8	C	Force exerted by spring = 5.0 N